

Ultrasonographic Appearance of Gallbladder Neoplasia in 14 Dogs and 1 Cat

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Abstract

Ultrasonography is the most commonly used diagnostic imaging tool for gallbladder disease in veterinary medicine. Primary gallbladder neoplasia is an uncommon finding with variable prognosis for which no studies have been published describing their ultrasonographic appearance and diagnosis. This retrospective, multicenter, case series study examines the ultrasonographic appearance of gallbladder neoplasia with histologically or cytologically confirmed diagnoses. A total of 14 dogs and 1 cat were analyzed. All discrete masses were sessile in shape and varied in size, echogenicity, location, and gallbladder wall thickening. All studies with images showing Doppler interrogation exhibited vascularity. Cholecystoliths were an uncommon finding, being present in only one case in this study, unlike in humans. The final diagnosis of the gallbladder neoplasia was neuroendocrine carcinoma (8), leiomyoma (3), lymphoma (1), gastrointestinal stromal tumor (1), extrahepatic cholangiocellular carcinoma (1), and adenoma (1). Findings from this study indicate that primary gallbladder neoplasms have variable sonographic appearances and cytologic and histologic diagnoses.

KEYWORDS

extrahepatic disease, canine, feline, masses, neoplasia, gallbladder

1 | INTRODUCTION

Gallbladder neoplasia is a rare and poorly documented cause of extrahepatic biliary disease in dogs and cats. In humans, adenocarcinoma of the gallbladder is the most common primary neoplasm of the biliary tree and the fifth most common tumor of the GI tract.¹ Other reported primary gallbladder neoplasms in humans include lymphoma² and sarcoma.³ Cholecystoliths are the most common concurrent abnormality in humans, occurring in 61% of humans diagnosed with gallbladder carcinoma.⁴ In dogs and cats the most commonly reported gallbladder neoplasm is neuroendocrine carcinoma.^{5,6,7,8} Other gallbladder neoplasms that have been reported in the dog include adenocarcinoma⁹ and leiomyoma.¹⁰ In both humans and dogs, the most common non-neoplastic mass in the gallbladder is the gallbladder polyp.^{1,11}

Ultrasonography, CT, and/or MRI are recommended in the diagnosis and identification of primary gallbladder neoplasia in humans.^{1,12,13} While radiographic contrast cholangiography can be performed in humans, it provides less information than ultrasound or CT.¹² Ultrasound findings of gallbladder neoplasia have been divided into two major subcategories in human literature, a distinct gallbladder mass lesion, and a mass that totally effaces the gallbladder.⁴ While ultrasound is the most common modality used in humans to detect gallbladder neoplasia, CT is recommended for early lesion detection, presence of metastases, and to assess for local infiltration in masses that are deemed surgically resectable on ultrasound.¹⁴ In humans, primary gallbladder adenocarcinoma has historically low survival rates, but advances in pre-operative diagnosis and staging have allowed for an overall increase in survival rates in a subset of patients that undergo surgery.¹⁵ Adenocarcinomas of the biliary tract in dogs have also often

metastasized by the time of diagnosis, with distant metastasis seen in 21 of 24 dogs with this diagnosis.⁹ For some patients with primary neuroendocrine carcinoma of the gallbladder, however, surgical removal of the gallbladder can be curative.⁵ Early diagnosis of gallbladder neoplasia is therefore critical to maximize treatment options and prognosis.

While isolated case reports exist,^{7,10,16-19} based on the authors' review of the literature, no study has been published describing the ultrasonographic characteristics of gallbladder neoplasia in dogs or cats. This study seeks to report on the ultrasonographic imaging findings seen in cytologically or histologically confirmed gallbladder neoplasms in dogs and cats.

2 | METHODS

2.1 | Selection and description of subjects

In this retrospective, multi-center case series study, patients with gallbladder neoplasia with ultrasound exams performed by Diplomates of the American College of Veterinary Radiology (ACVR) or the European College of Veterinary Diagnostic Imaging (ECVDI) and a confirmed diagnosis (by histopathology or cytology) were requested via emailing the ACVR.org Discourse and by emailing individual radiologists. Five institutions responded to this request with cases that are included in this study. Inclusion criteria included ultrasound images of the gallbladder, an ultrasonographic report by an ACVR or ECVDI-certified veterinary radiologist, and a histologic or cytologic diagnosis. Cases were excluded from the study if a definitive diagnosis of the gallbladder mass or ultrasonographic images of the gallbladder were not available. Approval by our institution's Animal Care and Use Committee was not required due to the retrospective design of the study. All decisions for inclusion or exclusion of cases were made by a diagnostic imaging intern with 2 years of post-graduate training experience.

2.2 | Data Recording and Analysis

The signalment (breed, age, sex), presenting clinical signs, history, and bloodwork were cataloged, as well as the surgical or necropsy report when available. All ultrasound images were reviewed by two ACVR-certified veterinary radiologists. The presence of a mass, shape of the mass (sessile or pedunculated), echogenicity of the mass in comparison to the liver, presence of vascularity if Doppler images were available, presence of subjective gallbladder wall thickening unassociated with any detected mass, presence of gallbladder "sludge", and presence of cholecystoliths were noted.

3 | RESULTS

Cases spanned from 2006 to 2021 and included 14 dogs and one cat. The patients ranged from six to 15 years of age (mean of 10.7 years,

median of 11 years), and included five spayed females, eight neutered males, and two intact males. Breeds consisted of two French bulldogs, two Brittany spaniels, one Irish wolfhound, one toy poodle, one boxer, one Yorkshire terrier, one American Eskimo dog, one Boston terrier, one golden retriever, one bulldog, one poodle, and one Australian shepherd mix. The single cat was a Maine coon.

Each individual case had one gallbladder neoplasm. Eight cases were diagnosed as neuroendocrine carcinoma (Figure 1,2,3), three as leiomyoma, one as lymphoma, one as a metastatic gastrointestinal stromal tumor (GIST), one as adenoma (Figure 4), and one as an extrahepatic cholangiocellular carcinoma (Figure 5). The single cat was diagnosed with extrahepatic cholangiocellular carcinoma (Figure 5). Three cases were diagnosed by cytology and twelve were diagnosed by histopathology. The diagnoses, breeds, ages, and sex of the patients are summarized in Table 1.

Images reviewed in this study were acquired with varying ultrasound machines (Logiq S8 Vet, GE, Boston MA; Aplio 500, Toshiba, Tokyo, Japan; Philips iU22, GE, Boston MA; Philips Epic 7G, GE, Boston MA; and Aplio MX, Toshiba, Tokyo, Japan). Either 5–8 MHz or 3–10 MHz microconvex transducers were used to obtain images. Patients were imaged in lateral or dorsal recumbency and probes were positioned in sagittal and/or transverse planes. Mode and patient preparation parameters were not recorded.

Ultrasound findings for all gallbladder neoplasms are summarized in Table 2. Details for each case are available in Supplement 1. All discrete masses that were detected in the provided images were sessile in shape. One gallbladder mass was not identified in the available ultrasound images, but had a description provided by the referring institution. Three cases did not exhibit a discrete mass; one case of neuroendocrine carcinoma was described as regular diffuse wall thickening (Figure 2), and two cases, one of adenoma (Figure 4) and one of neuroendocrine carcinoma exhibited focal irregular wall thickening. Four neoplasms were misdiagnosed in the original reports by the referring institutions, however were confirmed as neoplasia with histopathology; one of these was adenoma (Figure 4) and three were neuroendocrine carcinoma (Figure 2,3). Ultrasound images with Doppler were available for ten cases, all of which showed vascularity of the mass. Five cases had thickening of the gallbladder wall in a location not directly associated with or adjacent to the mass. The position of the mass within the gallbladder was able to be determined in eleven cases.

Measurements of the mass were available for ten cases, which ranged from 0.85cm to 4.5cm in maximum diameter with a median of 1.75cm. In the three cases without discrete masses, measurements of the gallbladder wall thickening were available and ranged from 0.2cm to 1.29cm. Regional lymphadenopathy was not captured in the images provided for this study, however two of the provided ultrasound reports did mention enlarged or irregular lymph nodes - one case of lymphoma had enlarged hepatic lymph nodes that measured up to 2.1cm in thickness and one case of metastatic GIST had a mildly enlarged pancreaticoduodenal lymph node that measured 0.4cm in thickness. No cases had evidence of biliary obstruction or abnormalities in gallbladder size. One case with regional peritoneal effusion and steatitis described in the original ultrasound report exhibited

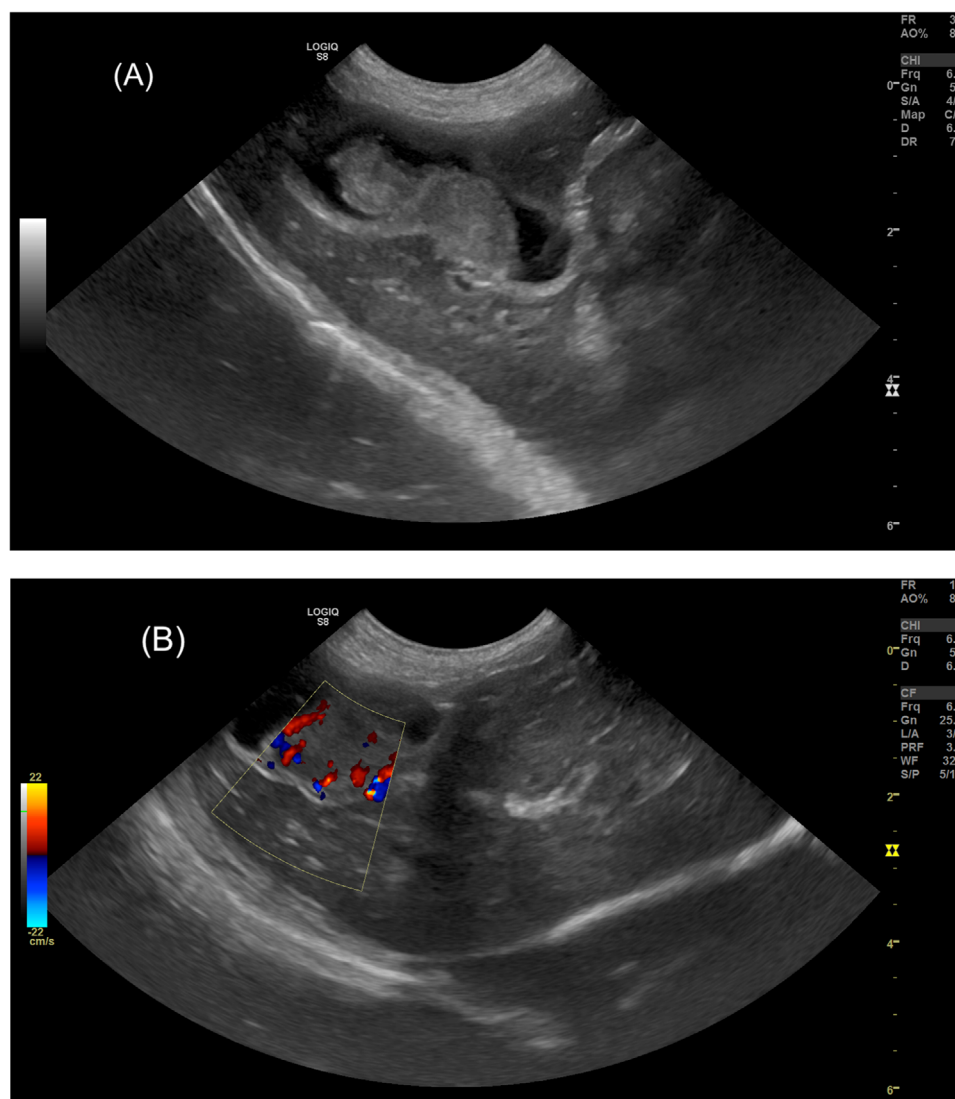


FIGURE 1 A, Sagittal ultrasonographic image of a dog diagnosed with neuroendocrine carcinoma obtained with an available ultrasound system (Logiq S8 Vet, GE, Boston MA) and an 10-MHz microconvex transducer. The mass is homogeneously hyperechoic, sessile in shape, arises from the dorsal gallbladder wall, and has no concurrent gallbladder wall thickening. The mass measures 1.7cm in greatest dimension. B, Transverse ultrasonographic image of the same dog obtained with an available ultrasound system (Philips Logiq S8 Vet, GE, Boston MA) and a 10-MHz microconvex transducer. Interrogation with color Doppler shows vascularity within the mass, differentiating the mass from avascular gallbladder “sludge”. [Color figure can be viewed at wileyonlinelibrary.com]

necrotizing and suppurative cholecystitis as well as neuroendocrine carcinoma on histopathology.

Only one case of gallbladder neoplasia was a metastasis from a previously resected GI stromal tumor, while the remaining cases were primary gallbladder neoplasms. The metastatic GI stromal tumor was the only case with distant metastasis and involved the spleen. One case of neuroendocrine carcinoma invaded the liver, and this case resulted in gallbladder rupture presumably secondary to the neoplasia. This case was presumed to arise from the gallbladder due to its histopathologic characteristics, but pathologists could not completely rule out hepatic origin due to the degree of invasion of hepatic tissue. No other primary gallbladder neoplasia showed evidence of distant or local metastasis.

Hepatobiliary bloodwork values (ALT, AST, GGT, total bilirubin) were at least partially available for fourteen patients, and of these, eleven patients had elevations of at least one value. Of the ten cases with bilirubin values available, four were elevated, and six of the nine cases with available GGT were elevated. Five of the 15 patients presented for unrelated diseases including pancreatitis, acute kidney injury, tachypnea/dyspnea, and neurologic disease. Seven patients had GI signs (anorexia, vomiting, and/or diarrhea) and concurrent elevated liver enzymes. Of the remaining three cases, one was receiving an ultrasound for routine staging following previous GIST removal of the GI tract, one was evaluated following palpation of an abdominal mass, and one had no clinical signs but was evaluated due to elevated liver enzymes discovered during routine bloodwork.



FIGURE 2 Transverse ultrasonographic image of a dog diagnosed with neuroendocrine carcinoma obtained with an available ultrasound system (Philips Epic 7G, GE, Boston MA) and an 8-MHz microconvex transducer. There is homogeneously hyperechoic diffuse, regular gallbladder wall thickening measuring up to 1.29 cm in greatest thickness. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/vim.13227)]



FIGURE 3 Sagittal ultrasonographic image of a dog diagnosed with neuroendocrine carcinoma obtained with an available ultrasound system (Philips Epic 7G, GE, Boston MA) and an 8-MHz microconvex transducer. The mass is heterogeneously hyperechoic and occupies the entire gallbladder lumen, measuring 4.5 cm in greatest dimension.



FIGURE 4 Transverse ultrasonographic image of a dog diagnosed with adenoma obtained with an available ultrasound system (Philips iU22, GE, Boston MA) and an 8-MHz microconvex transducer. The gallbladder has an irregularly thickened, heterogeneously hyperechoic wall that measures up to 0.2cm. The neoplasia was misdiagnosed by the performing ultrasonographer as an immature mucocele. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/vim.13227)]



FIGURE 5 Sagittal ultrasonographic image of a cat diagnosed with extrahepatic cholangiocellular carcinoma obtained with an available ultrasound system (Philips iU22, GE, Boston MA) and an 8-MHz microconvex transducer. The mass is homogeneously hyperechoic, sessile in shape, occupies nearly the entire lumen, and has concurrent gallbladder wall thickening. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/vim.13227)]

4 | DISCUSSION

In 14 dogs and one cat with gallbladder neoplasia identified with ultrasonography, masses had a variable appearance in echogenicity, location, and size. Factors not directly involving the mass, such as

presence of gallbladder “sludge” and the presence of gallbladder wall thickening in areas not associated with the mass, also varied. All discrete masses were sessile in shape. This is in contrast to the polypoidal appearance of gallbladder neoplasia most often described in humans, and may be attributed to the differences in types of neoplasia

TABLE 1 Representative breeds, age ranges, sex, and presenting signs for each neoplasia category

Neoplasia (N)	Breed (N)	Age (years +/- SD)	Sex (M,F)	Presenting Signs
Neuroendocrine Carcinoma (8)	French Bulldog (2)	10.3 +/- 1.5	MN (6)	GI Signs (4)
	Boxer (1)		MI (1)	Pancreatitis (1)
	American eskimo dog (1)		FS (1)	↑ liver enzymes; asymptomatic (1)
	Boston terrier (1)			Neurological workup (1)
	Golden Retriever (1)			Pneumonia (1)
	Bulldog (1)			
	Australian shepherd (1)			
Leiomyoma (3)	Brittany Spaniel (2)	12.3 +/- 1.2	MN (1)	Dyspnea (2)
	Yorkshire terrier (1)		FS (2)	Acute kidney injury (1)
Lymphoma (1)	Irish Wolfhound (1)	6	MI (1)	GI Signs (1)
GI Stromal Tumor (1)	Toy Poodle (1)	13	MN (1)	Previous GIST monitoring (1)
Cholangiocellular carcinoma (1)	Maine coon (1)	8	FS (1)	GI Signs (1)
Adenoma (1)	Poodle (1)	15	FS (1)	GI Signs (1)

Abbreviations: GI Signs, Gastrointestinal signs including vomiting, diarrhea, and/or inappetence; MN, male neutered; MI, male intact; FS, female spayed.

in humans as compared to dogs and cats.^{4,20} This is also helpful in distinguishing gallbladder neoplasia from polyps, which are typically pedunculated in dogs.¹¹ Given that suspected gallbladder polyps are typically not sampled for cytology or histology, however, this study may be lacking representation of neoplasms that have a more polyp-like, pedunculated appearance.

In this sample, the most common primary gallbladder neoplasm was neuroendocrine carcinoma, which constituted eight of the fifteen (53.3%) cases. This is in contrast to primary human gallbladder neoplasia, which is most commonly adenocarcinoma.²¹ Leiomyoma was the next most common primary gallbladder neoplasm in our study, constituting three of the 15 (20%) cases. Leiomyoma of the gallbladder has been previously reported in one dog,¹⁰ and a few case reports have been published in humans.^{22,23,24} Gallbladder lymphoma was found in one dog in our study, and has been reported in single case reports in the dog²⁵ and cat.²⁶ This has also been a reported gallbladder neoplasm in humans.²⁷ Although gallbladder adenomas were reported in a separate study,²⁸ this is only the second published case containing an ultrasonographic description in the veterinary literature.¹⁷

This is the first published report of gastrointestinal stromal tumor (GIST) metastasis to the gallbladder in dogs. Metastatic GIST to the gallbladder has been rarely reported in humans.²⁹ This case was initially diagnosed as neuroendocrine carcinoma on cytology with a second diagnosis of stromal tumor on histopathology, presumed to be metastasized from a previously resected intestinal GIST. This case emphasizes the difference in accuracy between cytology and histopathology of neoplasia in dogs,³⁰ however further investigation is necessary to determine the accuracy of cytology for gallbladder masses in dogs.

Four neoplasms were misidentified in the initial exam by the referring institution. Two of the neoplasms that were misidentified in the initial exam by the referring institution exhibited irregular wall thickening rather than a more typical mass-like appearance; one was diagnosed

as neuroendocrine carcinoma and one was diagnosed as adenoma on histopathology (Figure 4). One mass was misdiagnosed as a hepatic mass as it effaced the gallbladder lumen; this was diagnosed as neuroendocrine carcinoma arising from the gallbladder on histopathology (Figure 3). The fourth mass that was misdiagnosed as possible organized debris exhibited hyperechoic thickening of the gallbladder wall without reported evidence by the performing ultrasonographer of vascularity on Doppler; this was diagnosed on histopathology as neuroendocrine carcinoma (Figure 2). The more atypical appearance of these masses may account for the lack of identification, which also may lead to an underrepresentation of masses with irregular appearances in this study.

Doppler is used in veterinary medicine to distinguish vascular from avascular tissue, and has been reported as a primary method in which to differentiate a gallbladder mass/neoplasm from inspissated bile or a mucocele.⁷ In this study, images with Doppler interrogation were available in only ten of the fifteen cases, all of which showed intense vascularity (Figure 1B). One of the cases without Doppler images available mentioned lack of vascularity on Doppler in the initial report- due to this finding the conclusion of the performing ultrasonographer was that the structure may have been organized debris rather than a mass. The type of Doppler used by performing institutions varied, to include power and color Doppler, which may have played a role in detection of vascularity within the masses.

Factors associated with gallbladder malignancy in humans are size of the mass, age of the patient, presence of cholecystoliths, evidence of invasion to the liver, and gallbladder wall thickening.³¹ In this study, masses caused by malignant and benign neoplasia were variable in their sizes with overlap in their size ranges. No clear difference is seen in size between malignant and benign masses, although all four masses measuring >2cm in diameter were malignant (GIST, neuroendocrine carcinoma). The only masses that occupied the entirety of the gallbladder lumen were malignant, one case of neuroendocrine carcinoma and

TABLE 2 Ultrasound features found for each neoplastic diagnosis

Ultrasound characteristics	Neuroendocrine carcinoma	Leiomyoma	Lymphoma	Cholangiocellular Carcinoma	GI Stromal Tumor	Adenoma
Neoplasia visible on provided images						
Yes	8/8 (100%)	2/3 (67%)	1/1(100%)	1/1 (100%)	1/1 (100%)	1/1 (100%)
No	0/8 (0%)	1/3 (33%)	0/1 (0%)	0/1 (0%)	0/1 (0%)	0/1 (0%)
Discrete Mass	6/8 (75%)	2/2 (100%)	1/1 (100%)	1/1 (100%)	1/1 (0%)	0/1 (0%)
Wall thickening only	2/8 (25%)	0/2 (0%)	0/1 (0%)	0/1 (0%)	0/1 (0%)	1/1 (100%)
Maximal Recorded dimension (cm +/- SD)	2.5 +/- 1.5	1.7 +/- 0.3	1.3	0.85	1.79	0.2
Shape						
Sessile	6/6 (100%)	2/2 (100%)	1/1 (100%)	1/1 (100%)	1/1 (100%)	N/A
Irregular thickening	1/2 (50%)	N/A	N/A	N/A	N/A	1/1 (100%)
Regular thickening	1/2 (50%)	N/A	N/A	N/A	N/A	0/1 (0%)
Echogenicity						
Hyperechoic to liver	6/8 (75%)	2/3 (67%)	0/1 (0%)	1/1 (100%)	1/1 (100%)	1/1 (100%)
Isoechoic to liver	2/8 (25%)	0/3 (0%)	1/1 (100%)	0/1 (0%)	0/1 (0%)	0/1 (0%)
Hypoechoic to liver	0/8 (0%)	1/3 (33%)	0/1 (0%)	0/1 (0%)	0/1 (0%)	0/1 (0%)
Echotexture						
Heterogeneous	3/8 (38%)	0/2 (0%)	0/1 (0%)	0/1 (0%)	1/1 (100%)	1/1 (100%)
Homogeneous	5/8 (63%)	2/2 (100%)	1/1 (100%)	1/1 (100%)	0/1 (0%)	0/1 (0%)
GB wall thickening present						
Yes	2/8 (25%)	0/3 (0%)	1/1 (100%)	1/1 (100%)	0/1 (0%)	1/1 (100%)
No	6/8 (75%)	3/3 (100%)	0/1 (0%)	0/1 (0%)	1/1 (100%)	0/1 (0%)
Sludge present						
Yes	7/8 (88%)	2/3 (67%)	1/1 (100%)	1/1 (100%)	0/1 (0%)	1/1 (100%)
No	1/8 (13%)	1/3 (33%)	0/1 (0%)	0/1 (0%)	1/1 (100%)	0/1 (0%)
Choleliths present						
Yes	0/8 (0%)	1/3 (33%)	0/1 (0%)	0/1 (0%)	0/1 (0%)	0/1 (0%)
No	8/8 (100%)	2/3 (67%)	1/1 (100%)	1/1 (100%)	1/1 (100%)	1/1 (100%)
Location of lesion						
Wall	3/6 (50%)	0/3 (0%)	0/1 (0%)	0/1 (0%)	0/1 (0%)	N/A
Apex	0/6 (0%)	2/3 (67%)	1/1 (100%)	0/1 (0%)	1/1 (100%)	N/A
Neck	1/6 (17%)	1/3 (33%)	0/1 (0%)	0/1 (0%)	0/1 (0%)	N/A
Entire lumen/diffuse	2/6 (33%)	0/3 (0%)	0/1 (0%)	1/1 (100%)	0/1 (0%)	N/A

one case of cholangiocarcinoma. The age of patients with benign and malignant masses also overlapped. Of the four cases with gallbladder wall thickening, four were malignant and one was benign. Gallbladder wall thickening has been seen with infiltrative neoplasia^{21,32} or can be seen with cholecystitis,^{32,33} edema,^{32,34} and infarction.^{32,33} The only patient with choleliths in this study was diagnosed with leiomyoma of the gallbladder, a benign neoplasm. Choleliths are overall much less prevalent in dogs than in humans, with under 1% of hospitalized dogs showing this finding compared to 25% of the human population,³⁵ which may contribute to their low incidence within this study. While this is in contrast to humans, the lack of choleliths

in most patients in this study suggests that they are not a common characteristic of gallbladder neoplasia in dogs or cats.

Not all of the cases in this study initially presented for symptoms related to hepatic or extrahepatic disease. These findings indicate that gallbladder neoplasms may be subclinical, particularly if they are not resulting in biliary obstruction, rupture, or regional invasion or hemorrhage. Thorough investigation of the gallbladder is warranted even without clinical suspicion of disease.

12 of the 15 cases had concurrent gallbladder "sludge." While gallbladder sludge has been found to be associated with clinical disease in humans,³⁶ in dogs no association has been found between gallbladder

sludge and hepatobiliary disease.³⁷ The single cat in this study also had gallbladder sludge, which has been associated with liver disease in cats.³⁸

In conclusion, gallbladder neoplasia should be considered as a differential diagnosis for dogs or cats with any of the variable ultrasonographic appearances described in this paper, and with or without clinical signs of hepatobiliary disease. We did not identify any ultrasonographic features which would allow differentiation of histopathologic type for gallbladder neoplasms. Authors recommend investigation with Doppler for differentiating masses from organized or inspissated sludge.

LIST OF AUTHOR CONTRIBUTIONS

Category 1

- a) Conception and design: MacLeod, Reichle
- b) Acquisition of data: MacLeod, Cohen, Artiles, Fulkerson, Kurihara, Mattoon
- c) Analysis and Interpretation of Data: MacLeod, Reichle, Szabo

Category 2

- a) Drafting the article: MacLeod
- b) Revising article for intellectual content: MacLeod, Reichle, Szabo, Cohen, Artiles, Fulkerson, Kurihara, Mattoon

Category 3

- a) Final approval of the completed article: MacLeod, Reichle, Szabo, Cohen, Artiles, Fulkerson, Kurihara, Mattoon

Category 4

- a. Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved: MacLeod, Reichle, Szabo, Cohen, Artiles, Fulkerson, Kurihara, Mattoon

CONFLICT OF INTEREST

The authors have no conflict of interest

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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